MIL-R-52899B(ME)
8 July 1986
SUPERSEDING
MIL-R-52899A(ME)
30 August 1978

MILITARY SPECIFICATION

RAMP: LOADING VEHICLE

This specification is approved for use within the USA Belvoir Research, Development, and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a lightweight portable loading ramp which is used in conjunction with 4000-pound capacity, rough terrain and depot forklift trucks for loading and unloading between the ground and the floors of MILVANS and refrigerated containers.

1.2 Classification. Portable loading ramps shall be the following types, as specified (see 6.2).

Type I - Aluminum with aluminum grating.
Type II - Aluminum with steel grating.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel, and Seals.
RR-C-271	- Chains and Attachements, Welded and Weldless.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.

MILITARY

MIL-P-116	- Preservation Methods of.
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-G-3859	- Grease Guns, Hand-Operated, Lever, Push and Screw Type.
MIL-T-22085	- Tape, Adhesive, Preservation and Sealing.

STANDARDS

FEDERAL

FED-STD-H28	- Screw-Thread Standards for Federal Services.
FED-STD-595	- Colors.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-209	- Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-889	- Dissimilar Metals.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

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AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/AWS D1.1 Structural Welding Code - Steel, Section 5, Qualification.
ANSI/AWS D1.2 Structural Welding Code - Aluminum, Section 5, Qualification of Procedures and Personnel.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The vehicle loading ramp, hereinafter referred to as "ramp", shall be a basic aluminum structure with steel or aluminum grating as specified (see 1.2), and shall have a manually operated hydraulic system for vertical positioning. The ramp shall be equipped with a carrier assembly and detachable towbar, both of which may be steel construction.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3). Any changes or deviations of ramps from the approved first article during production will be subject to the approval of the contracting officer. Approval of the first article will not relieve the contractor of his obligation to furnish ramps conforming to this specification.

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3.3 Materials. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Material deterioration prevention and control. The ramp shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operation and storage environment to which the ramps may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.3.2 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the ramps may be newly fabricated from recovered materials to the maximum extent practicable, provided the ramps produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the ramps.

3.4 Design and construction. The ramp shall be designed to perform in accordance with the requirements of 3.7 for carrier floor heights ranging from 45 to 65 inches above ground level. The level-off section of the ramp shall be designed to preclude a crown on the deck and shall be in a level position when the lip is at a height of 62 inches +1-1/2 inches. The overall length of the ramp shall be 36 feet including a level-off section of at least 6 feet. The overall width of the ramp shall be not more than 96 inches with a usable width of not less than 88 inches except as specified herein. The upper end of the ramp shall have a solid lip of not less than 14 inches to support the ramp on the carrier floor. The overall width of the lip for the ramp shall be not more than 87 inches with a usable width of 84 inches. The ramp, from a point no more than 36 inches from the forward edge of the lip, shall be tapered to the rear edge of the lip. The gross weight of the ramp shall be not more than 6000 pounds. The ramp shall be equipped with handholds for positioning by four people manually to other location over semi-improved surfaces. The ramp shall be equipped with tow safety chains with grabhooks to secure the ramp to the carrier or dock (see 3.8). The ramp shall be equipped with side mounted reflectors near each end. The ramp design shall provide for safety curbs to prevent run-off of operating equipment while on the ramp. The safety curbs shall have a height of not less than 12 inches. The contour of the curb shall be essentially vertical and shall slope no more than 1 inch in 12 inches of height. The top edges of the curb shall have a radius of not less than .50 inch. The ramp shall have a rated capacity of 16,000 pounds. The wheels for moving the ramp shall be located approximately at the center of balance under-

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neath the ramp and shall be connected to the hydraulic actuating system so that the ramp can be utilized through the full height range without the wheels acting as load supports.

3.4.1 Deck. The approach at both ends of the deck shall consist of an integral safety tread plate. The approach edges of the tread plates shall be beveled. All other deck surfaces shall consist of open-type grating. The grating shall be an "all weather, self-cleaning, and draining type", with the top surface fully serrated to provide traction for forklift truck tires under all weather conditions. The grating holddown fastener shall not project above the top surface of the grating.

3.4.2 Hydraulic system. Vertical positioning of the ramp through a range from 45 to 65 inches above ground level shall be accomplished by a manually operated hydraulic system. The system shall be capable of elevating the ramp through this range within 150 seconds. The hydraulic system shall incorporate a means for filling and draining and an overload relief valve to protect the hydraulic system from damage due to overload. The drain shall have provision for drainage into a vessel. The pump shall be recessed and located in a protected position. The pump lever socket shall be positioned for ease of access and operation. The force required at the grip end of the pump lever to operate the pump against relief pressure shall be not more than 25 pounds.

3.4.2.1 Cylinders. Hydraulic system cylinders shall have an exclusion device to prevent foreign material and fluids from entering around the piston rod. The rod shall be hard chrome plated to prevent corrosion. After plating, maximum rod surface roughness shall be not more than 32 micro-inches.

3.4.3 Wheels. Wheels shall be industrial or automotive type and shall be mounted on tapered roller bearings. Seals shall be provided to retain grease and prevent entry of water or contaminants.

3.4.4 Tires. Each wheel shall be provided with a pneumatic tire. Tire loading shall not exceed the tire manufacturer's rating for the application. The recommended tire pressure shall be stenciled in 1-inch high (minimum) characters near the tires and labeled: "TP___PSI".

3.4.5 Suspension system. A coil or air spring suspension system shall be provided to permit towing of the ramp at speeds up to and including 25 miles per hour (see 3.7.4). Coil springs shall be pin-connected to permit placement in the towing position and in the operating position by one man without the use of tools. Means shall be provided to retain the springs clear of the ground in the operating position. Air springs shall be attached to the ramp and shall not require repositioning for the towing or operating positions. A resilient pad(s) shall be provided to prevent bottoming of the air spring.

3.4.6 Tow bar. The ramp shall be equipped with a detachable tow bar for towing. The tow bar shall be equipped with a quick-lock device for attachment to the ramp. The device shall be not separable from the towbar without the use of hand tools. The towing eye of the bar shall have an inside diameter not less than 3 inches and the center of the towing eye shall extend a distance of not

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less than 48 inches from the edge of the ramp. With the towing eye at a height of 28 inches, the ramp shall be approximately level. The tow bar shall be stowed beneath the ramp where it shall be easily accessible.

3.4.7 Lubrication. All surfaces requiring lubrication shall be provided with a suitable means for lubricating.

3.5 Maintainability. All assemblies shall be accessible for servicing repair and replacement as appropriate. The number of different sizes of fasteners and plugs shall be held to a minimum.

3.5.1 Lubrication fittings. Lubrication fittings shall conform to SAE J534. Fittings shall be located in a protected position and shall be accessible to a hand-operated grease gun conforming to MIL-G-3859 with 10-inch flexible extension. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts.

3.5.2 Interchangeability. All parts from the same manufacturer shall be manufactured to such standards that will permit replacement or adjustment without modification of parts or total equipment.

3.5.3 Threads. Threads shall conform to the applicable requirements of FED-STD-H28.

3.5.4 Fastening devices. All nuts, bolts, pins, and similar parts shall be installed with adequate means for preventing them from loosening or becoming out of adjustment. All fastening devices subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently deformed.

3.5.5 Instruction plates. Each ramp shall be equipped with instruction plates or diagrams, including warnings and cautions, describing any special or important procedures to be followed in assembling, operating, or servicing the ramp. Plates shall be attached in a location where the plate shall be both visible and legible.

3.6 Transportability.

3.6.1 Slinging provisions. Permanently affixed slinging provisions that enable the ramp to be lifted in its normal travel position shall be provided and shall conform to MIL-STD-209. Slinging provisions may also be used as tiedowns when such provisions conform to MIL-STD-209.

3.6.2 Tiedown provisions. Permanently affixed tiedown provisions that permit fastening the ramp to the floor or deck of a transportation medium shall be provided and shall conform to MIL-STD-209. Tiedown provisions may also be used as slinging provisions when such provisions conform to MIL-STD-209.

3.7 Performance.

3.7.1 Structure. The ramp shall be capable of withstanding stresses up to 200 percent of rated capacity without permanent deformation or indications of material or workmanship defects when tested as specified in 4.5.2.2.

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3.7.2 Hydraulic system. The hydraulic system shall show no evidence of leakage other than wetting of the pump piston assembly, malfunction, downward drift of more than .50 inch, or permanent deformation when tested as specified in 4.5.2.3.

3.7.3 Tow bar strength. The tow bar, when attached to the ramp and raised to a height of 28 inches shall withstand a free-drop to a hard surface at ground level without permanent deformation or evidence of breakage. The connection of the tow bar to the ramp shall show no signs of loosening during towing operations.

3.7.4 Towing operation. The ramp shall be capable of being towed by a vehicle having a pintle height of 28 inches with no evidence of permanent deformation or failure when tested as specified in 4.5.2.4. When air spring suspension is provided, air loss during the test shall not exceed 5 percent.

3.7.5 Endurance. The ramp, in normal operating position and elevated to a height of approximately 65 inches, shall be capable of withstanding, without permanent deformation or any evidence of breakage or hydraulic leakage, 3000 cycles of operation when tested as specified in 4.5.2.5.

3.8 Safety. Safety chains shall be not less than 8 feet in length. Each chain shall be equipped with a grabhook on one end and a shackle on the opposite end. Chain, shackle, and hook shall conform to RR-C-271, type 1, grade C, class 4, minimum .25 inch nominal size, type IV, and type V, class 1 or 2, respectively.

3.8.1 Hazardous edges and surfaces. Edges and surfaces exposed to operating and maintenance personnel shall be smooth and rounded to the extent that a hazardous surface does not exist.

3.9 Detail requirements. When the type II ramp is specified, the steel grating shall be galvanized by the hot-dipped process or shall have two coats of baked epoxy base synthetic finish, or shall have two coats of urethane enamel with a 2.0 mil dry film thickness.

3.10 Color and finish. The safety curb, the outside of the ramp side beams and, when of steel, the carrier assembly and tow bar shall be cleaned, treated, and painted. The surfaces to be painted shall be thoroughly cleaned and shall be dry and free from mill scale, oil, grease, dirt, and rust. As soon as possible after cleaning, surfaces shall be treated with a pretreatment primer and two finish coats. Primer and finish coats shall be applied in accordance with normal commercial practices to assure complete coverage and durability of the finish. The finish coat, when dry, shall be a smooth, even surface, free from runs, sags, peels, chips, blisters, and areas of thin film or no film. Paint shall be kept off working surfaces when interference with the working parts will result. Finish coat for the curbs and side beams shall be a visibility yellow in accordance with FED-STD-595, color No. 13538. Finish coat for the carrier assembly and tow bar shall be black or silver in accordance with FED-STD-595; color No. 27038 or 17178, respectively.

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3.11 Identification marking. The ramp shall be identified in accordance with MIL-STD-130.

3.11.1 Identification plates. Each ramp shall be provided with an identification plate. The identification plate shall include manufacturer's name, type, model, serial number, and load capacity of the unit.

3.12 Workmanship. The ramp, when tested as specified herein, shall perform loading and towing operations without permanent deformation, breakage of connections, malfunction, or component interference. All parts of the rampshall be clean and free from sand, rust, dirt, fins, pits, sprues, scale, flux, and other harmful extraneous materials.

3.12.1 Rivet connections. Rivets shall fill the holes completely. The upset rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member, and shall be in accordance with SAE J492.

3.12.2 Bolted connections. Boltholes shall be accurately formed. Washers or lockwashers or a combination of the two shall be provided on all bolts, studs, and capscrews having straight thread. Self-locking nuts are acceptable in lieu of standard nuts and lockwashers. All fasteners shall be correctly tightened and nuts shall have full thread engagement.

3.12.3 Welding.

3.12.3.1 Aluminum. The aluminum surfaces of parts to be welded shall be free from scale, paint, grease, oxide film, and other foreign matter. The welding shall be done by the inert gas, shielded-arc method. Welds may be ground, filed, buffed, or chipped but shall not be hammered, and shall be free from discontinuities such as base metal cracks adjacent to or behind welds, cracks in weld metal, undercutting of base metal, lack of fusion with parent metal or between parts, lack of penetration, lack of prescribed fit, or spatter.

3.12.3.2 Steel. The steel surfaces of parts to be welded shall be free from rust, scale, paint, grease, mill scale which can be removed by chipping and wire breaking, and other foreign matter. Parent materials, weld filler metals, and fabrication techniques shall be as required to enable the ramp to conform to the examination and test requirements specified in section 4.

3.12.3.3 Welders. Before assigning any welder to manual welding work covered by this specification, the contractor shall obtain certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

ANSI/AWS D1.1 Structural Welding Code-Steel, Section 5, Qualification.
ANSI/AWS D1.2 Structural Welding Code-Aluminum; Section 5, Qualification of Procedures and Personnel.

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Certification shall be made available for review by the contracting officer or designated representative upon request. Contractors who make only horizontal welds need not qualify welders for "all positon welding".

3.12.4 Machine work. All parts shall be manufactured to insure accuracy of assembly through the use of correct jigs, fixtures, or tape-controlled machines, or any combination thereof.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)
c. Inspection of packaging (see 4.6)

4.3 First article inspection.

4.3.1 Examination. The first article ramp shall be examined for the defects specified in 4.5.1. Presence of one more defects shall be cause for rejection.

4.3.2 Tests. The first article ramp shall be subjected to the tests marked "X" in column 1 of table I in the order indicated in the table. Failure of any test shall be cause for rejection of the first article.

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TABLE I. Test schedule.

First Article	Quality Conformance	Test	Test Paragraph	Requirement Paragraph
1	2	3	4	5
X		Curb.	4.5.2.7	3.4
X		Slinging provisions.	4.5.2.8	3.6.1
X		Tiedown provisions.	4.5.2.8	3.6.2
X		Overload.	4.5.2.2	3.7.1
X	X	Hydraulic system.	4.5.2.3	3.7.2
X		Tow bar.	4.5.2.5	3.7.3
X		Towing operation.	4.5.2.4	3.7.4
X		Endurance.	4.5.2.6	3.7.5

4.4 Quality conformance inspection.

4.4.1 Examination. Each ramp shall be examined for the defects specified in 4.5.1. AQL shall be 1.0 percent defective for major defects and 4.0 percent defective for minor defects.

4.4.2 Tests. Each ramp shall be subjected to the test marked "X" in column 2 of table I, except that the 2-hour period to check drift and the force required to actuate the pump are not required. Failure of the test shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The ramp shall be examined as specified herein for the following defects.

Major

Defect No.	Defect	Requirement paragraph
101.	Materials not as specified.	3.3
102.	Materials not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration for the applicable storage and operating environments.	3.3.1

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103.	Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.	3.3.1.1
104.	Contractor does not have documentation available for identification of material, material finishes, or treatment.	3.3.1.2
105.	Used, rebuilt or remanufactured components, pieces, or parts incorporated in the ramp.	3.3.2
106.	Dimensions not as specified.	3.4
107.	Weight exceeds 6000 lbs.	3.4
108.	Handholds for positioning ramp not provided.	3.4
109.	Wheels act as load supports within 2-inch height range of 45 inches to 65 inches.	3.4
110.	Approach edges of tread plates not beveled.	3.4.1
111.	Grating is not all-weather, self-cleaning, and draining type.	3.4.1
112.	Grating holddown fasteners project above top surface.	3.4.1
113.	Pump not recessed in protected position.	3.4.2
114.	Pump lever socket not accessible.	3.4.2
115.	Cylinders not as specified.	3.4.2.1
116.	Wheels not as specified.	3.4.3
117.	Tire loading exceeds tire manufacturer's recommendations.	3.4.4
118.	Suspension not as specified.	3.4.5
119.	Means not provided to retain springs clear of the ground in operating position.	3.4.5
120.	Resilient pad not provided with air spring suspension.	3.4.5
121.	Air springs require repositioning.	3.4.5
122.	Quick-lock device separable from tow bar without the use of handtools.	3.4.6
123.	Tow bar storage not as specified.	3.4.6
124.	Lubrication means not provided as specified.	3.4.7
125.	Grease fittings not accessible.	3.4.7
126.	Assemblies not accessible for maintenance.	3.5
127.	Grease fittings not in accordance with SAE J534.	3.5.1
128.	Interchangeability not as specified.	3.5.2
129.	Screw threads not in accordance with FED-STD-H28.	3.5.3
130.	Fastening devices not as specified.	3.5.4
131.	Safety chains not as specified.	3.8
132.	Hazardous edges and surfaces not rounded and smooth.	3.8.1
133.	Type II ramp not finished as specified.	3.9
134.	Workmanship not as specified.	3.12
135.	Rivet connections not as specified.	3.12.1
136.	Bolted connections not as specified.	3.12.2
137.	Welding not as specified.	3.12.3, 3.12.3.1, 3.12.3.2

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138.	Machine work not as specified.	3.12.4
139.	Welder and welders certification not as specified	3.12.3.3

Minor

201.	Instruction plates not furnished or incomplete.	3.5.5
202.	Color and finish not as specified.	3.10
203.	Identification plate missing, incomplete, or illegible.	3.11.1

4.5.2 Tests.

4.5.2.1 Test conditions. Prior to test, the ramp shall be lubricated with oils and greases as recommended by the contractor. Tests shall be conducted in the order listed.

4.5.2.2 Overload. The assembled ramp shall be supported at each end in a horizontal position and subjected to a static load of 200 percent of the rated load. The load shall be applied through an area extending along and not more than 9 inches on each side of the lateral centerline of the ramp. The ramp wheels shall not act as load supports during this test. The ramp shall be left under this load for a period of 1 hour, after which the load shall be removed and the ramp inspected. Nonconformance to 3.7.1 shall constitute failure of this test.

4.5.2.3 Hydraulic. The ramp in normal operating position shall be lowered to a height of not more than 45 inches. Operate the hydraulic system manually, raising the lip to a height of not less than 65 inches in not more than 150 seconds. Continue operating the pump until relief pressure is reached. Measure the force, at the end of the handle, required to actuate the pump. Lower the ramp to its lowest position. Operate the hydraulic system through 10 cycles, each consisting of raising the ramp to its highest position and then lowering to its lowest position. After completion of the 10 cycles, all moving parts shall be examined for evidence of wear to the operating mechanism. The hydraulic components and lines shall be examined for leakage and damage. The ramp shall then be raised and held at the fully raised position for not less than 2 hours. There shall be no leakage and the ramp shall not drift downward more than .50 inch during this period. Nonconformance to 3.7.2 shall constitute failure of this test.

4.5.2.4 Towing. The ramp shall be coupled to a towing vehicle and shall be towed over a semi-improved road at an average speed of not less than 15 miles per hour (mph) for 14 hours with maximum speed of not less than 25 mph maintained for a distance of not less than 1 mile each hour. In addition, the ramp shall be towed cross-country, over rough terrain for a distance of not less than 50 miles at speeds up to 5 mph. When air spring suspension is furnished, inflation pressure shall be checked and recorded before starting the test and upon completion of the test. Nonconformance to 3.7.4 or evidence of contamination in wheel assemblies shall constitute failure of this test.

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4.5.2.5 Tow bar. With the tow bar attached and the ramp in towing position, raise the tow bar to a height of not less than 28 inches. Allow the ramp to fall free such that the tow bar strikes a hard surface at ground level. Nonconformance to 3.7.3 shall constitute failure of this test.

4.5.2.6 Endurance. The ramp, in operating position, shall be attached to a trailer-mounted container, loading dock, or equivalent facility. The lip of the ramp, after positioning, shall be at a height of not less than 60 inches. A forklift truck of not less than 4000-pound capacity and compatible with the ramp shall ascend the ramp from ground level driving onto the floor of the container, dock, or equivalent facility, deposit the load, descend the ramp to the ground level, ascend the ramp, pick up the load, and descend the ramp to ground level. This shall constitute one complete cycle of the test. The test shall be continued for a total of not less than 3000 cycles. Nonconformance to 3.7.5 shall constitute failure of this test.

4.5.2.7 Curb. A force of 3000 pounds shall be applied perpendicular to the vertical plan of the side truss over an area of 9 square inches, centered 2 inches below the top surface of the curb. The test shall be repeated with the direction of the force reversed. The force shall be applied at a minimum of three points along the length of the curb with one point located approximately at the center. The remaining two points shall be located each side of the center between any vertical supports of the curb. Any evidence of permanent deformation shall constitute failure of this test.

4.5.2.8 Transportability. Slinging and tiedown provisions shall be tested as specified in MIL-STD-209. Nonconformance to MIL-STD-209 shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection a completed pack prepared for shipment shall be considered a unit product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects:

No.	Defect	A	Comm
140.	Preservative application and selection not as specified.	5.1.1.1	5.1.2
141.	Unfinished surface not protected as specified.	5.1.1.2	
142.	Bearings not lubricated as specified.	5.1.1.3	
143.	Hydraulic system not preserved as specified.	5.1.1.4	

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No.	Defect	A	Comm
144.	Technical data not preserved as specified.	5.1.1.5	
145.	Consolidation not as specified.	5.1.1.5	
146.	Packing not as specified.	5.2.1	5.2.2
147.	Marking not as specified.	5.3.1	5.3.2

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Preservative application and selection. The provisions of MIL-P-116 shall apply when selecting and determining the applicability of preservatives. Preservatives specified herein by "P" numbers shall conform to the applicable specifications listed in MIL-P-116 and shall be applied in accordance with MIL-P-116.

5.1.1.2 Unfinished surfaces. Unprotected exterior metal surfaces of components requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with type P-1 preservative.

5.1.1.3 Bearings. Bearings shall be lubricated to capacity with the manufacturer's recommended lubricant containing a rust inhibitor.

5.1.1.4 Hydraulic system. The hydraulic fluid supply tank shall be filled to the operating level with the hydraulic fluid required for operation. The piston shall be retracted as far as practicable into the cylinder. When the piston cannot be fully retracted into the cylinder, the exposed portion of the piston shall be coated with type P-6 or P-11 preservative and the coated surface wrapped or covered with barrier material conforming to MIL-B-121, type II, grade A, class 2 extending the wrap approximately 2 inches on the ram cylinder. The barrier material shall be secured in place with tape conforming to MIL-T-22085, type II. The pump lever shall be secured in a manner to prevent movement.

5.1.1.5 Consolidation. The equipment manual and any accessories or installation hardware shall be consolidated together in a close-fitting box conforming to PPP-B-601, overseas type, grade B, style optional. The contents shall be blocked, braced, or cushioned as applicable within the box to prevent movement or damage.

5.1.2 Commercial. Each complete ramp shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A or commerical, as specified (see 6.2).

5.2.1 Level A. The ramp shall be prepared for unboxed (mobile) shipment. The consolidation box (see 5.1.1.5) shall be secured to the ramp in a protected location that will not interfere with the lifting or its mobility. The securing shall be accomplished by the use of steel strapping conforming to QQ-S-781, type

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I or IV, finish B, size as applicable. The tow bar shall be stowed and, if needed to prevent loss, secured in place with strapping specified herein.

5.2.2 Commercial. Each complete ramp shall be packed in accordance with ASTM D 3951.

5.3 Marking.

5.3.1 Military. Marking for military levels of protection shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Marking for commercial packaging shall be in accordance with ASTM D 3951. Additionally, each unit shall be marked with the gross weight and cube.

6. NOTES

6.1 Intended use. The lightweight portable ramps covered by this specification are used for loading and unloading with 4000-pound capacity rough terrain and depot forklift trucks between ground level and the floors of trailer mounted MILVANS and containers. Type I ramp with aluminum grating is lighter than the type II ramp with steel grating.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type required (see 1.2).
c. When a first article is not required for inspection and approval (see 3.2, 4.3 and 6.3).
d. Degree of preservation and degree of packing required (see 5.1 and 5.2).

6.3 First article. When a first article inspection is required, the items should be a preproduction model. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests, and approval of the first article test results and disposition of the first article.

6.4 Subject term (key word) listing.

Aluminum structure
Lightweight
Manual hydraulic vertical positioning
Portable
Ramp

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6.5 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - ME

Review activities:
Army - AT
DLA - GS

Preparing activity:
Army - ME

Project 3990-A186

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-R-52899B(ME)	Ramp, Loading Vehicle

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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